# Supplementary Material for: Knowledge-Guided Dual-Phase Swarm Evolutionary Learning for High-Dimensional Feature Selection

Anonymous Author(s)

---

**Abstract**

This supplementary document provides comprehensive extended experimental records, complete per-dataset metrics, detailed hyper-parameter sensitivity grids, convergence trajectories, and full statistical significance matrices supporting the main manuscript.

---

**Table of Contents**

- *Section S1*: Benchmark Dataset Characteristics and Selection Rationale
- *Section S2*: Comparison Algorithms and Detailed Hyper-Parameter Configurations
- *Section S3*: Complete Per-Dataset Classification Accuracy and Feature Compactness Metrics
- *Section S4*: Granular Per-Dataset Component Contribution (Ablation) Results (Table S4)
- *Section S5*: Granular Per-Dataset Computational Wall-Clock Runtime Measurements (Table S5)
- *Section S6*: Full Multi-Dataset Parameter Sensitivity Evaluation Grids (Table S6)
- *Section S7*: Full Convergence Trajectories and Boxplot Distributions
- *Section S8*: Extended Statistical Significance Testing Matrices

## Section S1: Benchmark Dataset Characteristics and Selection Rationale

Table 2 summarizes the statistical characteristics of the 20 benchmark datasets investigated in this study, spanning biomedical gene-expression microarrays, high-dimensional text categorization, visual object recognition, and acoustic phoneme analysis.

Table 1:

Table 2: Characteristics of the 20 benchmark datasets.

| No. | Dataset | Samples ($n$) | Features ($D$) | Classes ($C$) | Domain |
|---|---|---|---|---|---|
| 1 | Adenoma | 36 | 7,457 | 2 | Gene expression |
| 2 | ALL_AML | 72 | 7,129 | 2 | Gene expression |
| 3 | ALL1 | 128 | 12,625 | 2 | Gene expression |
| 4 | ALL4 | 93 | 12,625 | 2 | Gene expression |
| 5 | Colon | 62 | 2,000 | 2 | Gene expression |
| 6 | DLBCL | 77 | 7,129 | 2 | Gene expression |
| 7 | Gastric | 65 | 22,645 | 2 | Gene expression |
| 8 | Gastric1 | 144 | 22,283 | 2 | Gene expression |
| 9 | Gastric2 | 124 | 22,283 | 2 | Gene expression |
| 10 | Leukaemia | 72 | 7,129 | 2 | Gene expression |
| 11 | Lymphoma | 45 | 4,026 | 2 | Gene expression |
| 12 | Myeloma | 173 | 12,625 | 2 | Gene expression |
| 13 | Prostate | 102 | 12,625 | 2 | Gene expression |
| 14 | SRBCT | 83 | 2,308 | 4 | Gene expression |
| 15 | T1D | 101 | 54,675 | 2 | Gene expression |
| 16 | BASEHOCK | 1,993 | 4,862 | 2 | Text |
| 17 | PCMAC | 1,943 | 3,289 | 2 | Text |
| 18 | COIL20 | 1,440 | 1,024 | 20 | Image |
| 19 | ISOLET | 1,560 | 617 | 26 | Speech |
| 20 | madelon | 2,600 | 500 | 2 | Synthetic |

*Note*: Datasets span $n \in [36, 2{,}600]$, $D \in [500, 54{,}675]$, and $C \in [2, 26]$.

## Section S2: Comparison Algorithms and Detailed Hyper-Parameter Configurations

Table 4 lists the full taxonomy and categorization of the 13 comparison algorithms evaluated in the comparative study, and Table 6 specifies the precise hyper-parameter configurations used across all algorithms to ensure strict reproducibility.

Table 3:

Table 4: Summary of the 14 compared methods investigated in this study.

| Reference | Method | Category |
|---|---|---|
| This work | KDS-EL | Proposed |
| This work | KDS-EL w/o TPS | Ablative variant |
| This work | KDS-EL w/o TPS&KER | Ablative variant |
| [1] | MEL | Multi-task swarm |
| [2] | FWPSO | Single-objective |
| [3] | PSO-CSM | Single-objective |
| [4] | SaWDE | Single-objective |
| [5] | IGPSO | Single-objective |
| [6] | BPSO | Single-objective |
| Self-built | W-MOPSO-FS | Multi-objective |
| [7] | fMOPSO-FS | Multi-objective |
| [8] | NSGA-II-FS | Multi-objective |
| [9] | MOEA-D-FS | Multi-objective |
| [10] | MOPSO-FS | Multi-objective |

## Section S3: Complete Per-Dataset Classification Accuracy and Feature Compactness Metrics

Tables 7 and 8 present the complete per-dataset 5-fold cross-validation classification accuracy (mean ± standard deviation) and average selected feature counts across all 20 benchmark datasets and all 14 evaluated algorithms.

Table 5:

Table 6: Hyper-parameter configurations and experimental settings.

| Parameter / Setting | Value / Configuration |
|---|---|
| *Common experimental setup:* | |
| Population size ($N$) | 20 |
| Maximum iterations ($T$) | 100 |
| $k$-NN classifier neighborhood ($k$) | 3 |
| Real-to-binary threshold ($\theta$) | 0.60 |
| Cross-validation scheme | 5-fold stratified, shuffled |
| Number of independent runs | 10 |
| *KDS-EL (full framework):* | |
| Inertia weight ($\omega$) | 0.90 |
| Acceleration coefficients ($c_1, c_2, c_3$) | 2.0, 2.0, 2.0 |
| Knowledge weight decay factor ($\gamma$) | 0.99 |
| Elite perturbation fraction ($\phi$) | 0.01 |
| Elite pool size ($k_e$) | 3 |
| Random exploration probability ($p_{\text{rand}}$) | 0.10 |
| MI-guided initialization ratio ($\rho$) | 0.50 |
| Stagnation patience threshold ($\tau$) | 20 generations |
| Phase split ratio ($\alpha$) | 0.70 |
| Phase-1 objective weights ($\mathbf{w}_1$) | [0.70, 0.30] |
| Phase-2 objective weights ($\mathbf{w}_2$) | [0.95, 0.05] |
| *Ablative configurations:* | |
| Fixed weights (KDS-EL w/o TPS) | $\mathbf{w}_s$ = [0.90, 0.10] |
| Parameters (KDS-EL w/o TPS&KER) | $\omega = 0.90,\ c_{1,2,3} = 2.0,\ \rho = 0.50,\ \tau = 20$ |
| Parameters (Standard MEL [1]) | $\omega = 0.90,\ c_{1,2,3} = 2.0,\ \mathbf{w}_s$ = [0.90, 0.10] |
| *Benchmark competitor algorithms:* | |
| Competitor algorithms [2, 3, 4, 5, 6, 7, 8, 9, 10] | Per original literature ($N = 20, T = 100$) |

Table 7: Comparison of classification accuracy (mean ± std) across 20 benchmark datasets.

| Dataset | KDS-EL | w/o TPS | w/o KER | MEL | FWPSO | PSO-CSM | SaWDE | IGPSO | BPSO | W-MPSO | fMPSO | NSGA-II | MOEA/D | MOPSO |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ALL1 | 0.9998±0.0001 | 0.9996±0.0000$^{\approx}$ | 0.9996±0.0000$^{+}$ | 0.9664±0.0023$^{+}$ | 0.9952±0.0015$^{+}$ | 0.9900±0.0002$^{+}$ | 0.9710±0.0011$^{+}$ | 0.9802±0.0002$^{+}$ | 0.9534±0.0003$^{+}$ | **1.0000±0.0000**$^{-}$ | **1.0000±0.0000**$^{-}$ | **1.0000±0.0000**$^{-}$ | 0.9985±0.0031$^{+}$ | **1.0000±0.0000**$^{-}$ |
| ALL4 | **0.9522±0.0061** | 0.9418±0.0113$^{\approx}$ | 0.9415±0.0112$^{+}$ | 0.8704±0.0661$^{+}$ | 0.8617±0.0304$^{+}$ | 0.8363±0.0135$^{+}$ | 0.8062±0.0217$^{+}$ | 0.7914±0.0179$^{+}$ | 0.7569±0.0069$^{+}$ | 0.8068±0.0047$^{+}$ | 0.8015±0.0089$^{+}$ | 0.7694±0.0118$^{+}$ | 0.7737±0.0184$^{+}$ | 0.7654±0.0074$^{+}$ |
| ALL_AML | **0.9995±0.0002** | 0.9990±0.0002$^{+}$ | 0.9886±0.0016$^{+}$ | 0.9539±0.0008$^{+}$ | 0.9850±0.0067$^{+}$ | 0.9748±0.0039$^{+}$ | 0.9591±0.0012$^{+}$ | 0.9669±0.0003$^{+}$ | 0.9426±0.0034$^{+}$ | 0.9943±0.0070$^{\approx}$ | 0.9900±0.0065$^{+}$ | 0.9857±0.0000$^{+}$ | 0.9857±0.0000$^{+}$ | 0.9857±0.0000$^{+}$ |
| Adenoma | **0.9996±0.0002** | **0.9996±0.0001**$^{\approx}$ | 0.9958±0.0081$^{+}$ | 0.9410±0.0011$^{+}$ | 0.9783±0.0143$^{+}$ | 0.9665±0.0108$^{+}$ | 0.9461±0.0014$^{+}$ | 0.9527±0.0003$^{+}$ | 0.9287±0.0002$^{+}$ | 0.9800±0.0131$^{+}$ | 0.9743±0.0086$^{+}$ | 0.9714±0.0000$^{+}$ | 0.9714±0.0000$^{+}$ | 0.9714±0.0000$^{+}$ |
| BASEHOCK | 0.9113±0.0120 | 0.8997±0.0114$^{+}$ | 0.8597±0.0062$^{+}$ | 0.8518±0.0063$^{+}$ | 0.8715±0.0095$^{+}$ | 0.8637±0.0050$^{+}$ | 0.8860±0.0065$^{+}$ | 0.8479±0.0062$^{+}$ | 0.8597±0.0036$^{+}$ | 0.8847±0.0066$^{+}$ | 0.8861±0.0045$^{+}$ | **0.9238±0.0082**$^{-}$ | 0.9150±0.0086$^{+}$ | 0.8920±0.0034$^{+}$ |
| COIL20 | 0.9939±0.0011 | 0.9919±0.0008$^{+}$ | 0.9619±0.0007$^{+}$ | 0.9634±0.0017$^{+}$ | 0.9894±0.0018$^{+}$ | 0.9782±0.0006$^{+}$ | 0.9763±0.0017$^{+}$ | 0.9707±0.0010$^{+}$ | 0.9573±0.0006$^{+}$ | 0.9960±0.0009$^{-}$ | 0.9960±0.0005$^{-}$ | **0.9971±0.0007**$^{-}$ | 0.9958±0.0010$^{-}$ | 0.9964±0.0009$^{-}$ |
| Colon | **0.9500±0.0075** | 0.9347±0.0094$^{+}$ | 0.9211±0.0092$^{+}$ | 0.8548±0.0276$^{+}$ | 0.9221±0.0203$^{+}$ | 0.8880±0.0090$^{+}$ | 0.8867±0.0110$^{+}$ | 0.8636±0.0101$^{+}$ | 0.8529±0.0059$^{+}$ | 0.9033±0.0072$^{+}$ | 0.8988±0.0071$^{+}$ | 0.8731±0.0097$^{+}$ | 0.8745±0.0145$^{+}$ | 0.8806±0.0104$^{+}$ |
| DLBCL | **0.9995±0.0002** | 0.9993±0.0004$^{\approx}$ | 0.9902±0.0016$^{+}$ | 0.9483±0.0058$^{+}$ | 0.9819±0.0092$^{+}$ | 0.9819±0.0048$^{+}$ | 0.9629±0.0055$^{+}$ | 0.9655±0.0063$^{+}$ | 0.9435±0.0069$^{+}$ | 0.9974±0.0052$^{\approx}$ | 0.9960±0.0061$^{\approx}$ | 0.9781±0.0060$^{+}$ | 0.9781±0.0060$^{+}$ | 0.9872±0.0058$^{+}$ |
| Gastric | **0.9782±0.0034** | 0.9677±0.0093$^{+}$ | 0.9748±0.0088$^{+}$ | 0.8955±0.0786$^{+}$ | 0.9191±0.0340$^{+}$ | 0.8874±0.0087$^{+}$ | 0.8234±0.0191$^{+}$ | 0.8564±0.0087$^{+}$ | 0.8046±0.0093$^{+}$ | 0.8908±0.0046$^{+}$ | 0.8846±0.0077$^{+}$ | 0.8154±0.0069$^{+}$ | 0.8169±0.0083$^{+}$ | 0.8215±0.0102$^{+}$ |
| Gastric1 | **0.9800±0.0003** | 0.9738±0.0004$^{+}$ | 0.9642±0.0025$^{+}$ | 0.9310±0.0155$^{+}$ | 0.9597±0.0032$^{+}$ | 0.9496±0.0024$^{+}$ | 0.9348±0.0030$^{+}$ | 0.9402±0.0018$^{+}$ | 0.9145±0.0002$^{+}$ | 0.9588±0.0021$^{+}$ | 0.9581±0.0000$^{+}$ | 0.9588±0.0021$^{+}$ | 0.9554±0.0034$^{+}$ | 0.9588±0.0021$^{+}$ |
| Gastric2 | **0.9981±0.0029** | 0.9936±0.0029$^{+}$ | 0.9872±0.0038$^{+}$ | 0.9522±0.0170$^{+}$ | 0.9837±0.0049$^{+}$ | 0.9710±0.0002$^{+}$ | 0.9532±0.0038$^{+}$ | 0.9623±0.0001$^{+}$ | 0.9357±0.0034$^{+}$ | 0.9837±0.0000$^{+}$ | 0.9838±0.0002$^{+}$ | 0.9765±0.0025$^{+}$ | 0.9765±0.0025$^{+}$ | 0.9832±0.0025$^{+}$ |
| ISOLET | 0.8894±0.0056 | 0.8898±0.0090$^{\approx}$ | 0.8557±0.0063$^{+}$ | 0.8501±0.0067$^{+}$ | 0.8822±0.0109$^{\approx}$ | 0.8584±0.0043$^{+}$ | 0.8903±0.0076$^{\approx}$ | 0.8468±0.0070$^{+}$ | 0.8607±0.0027$^{+}$ | 0.8883±0.0032$^{\approx}$ | 0.8874±0.0066$^{\approx}$ | **0.9090±0.0059**$^{-}$ | 0.8963±0.0059$^{-}$ | 0.8828±0.0036$^{+}$ |
| Leukaemia | **0.9995±0.0002** | 0.9993±0.0002$^{+}$ | 0.9895±0.0017$^{+}$ | 0.9534±0.0015$^{+}$ | 0.9886±0.0059$^{+}$ | 0.9775±0.0059$^{+}$ | 0.9587±0.0017$^{+}$ | 0.9682±0.0037$^{+}$ | 0.9416±0.0002$^{+}$ | 0.9886±0.0057$^{+}$ | 0.9886±0.0057$^{+}$ | 0.9857±0.0000$^{+}$ | 0.9857±0.0000$^{+}$ | 0.9857±0.0000$^{+}$ |
| Lymphoma | **0.9994±0.0003** | **0.9994±0.0004**$^{\approx}$ | 0.9868±0.0102$^{+}$ | 0.8731±0.0239$^{+}$ | 0.9404±0.0306$^{+}$ | 0.9206±0.0098$^{+}$ | 0.8841±0.0092$^{+}$ | 0.8861±0.0139$^{+}$ | 0.8585±0.0081$^{+}$ | 0.9178±0.0102$^{+}$ | 0.9200±0.0109$^{+}$ | 0.8844±0.0133$^{+}$ | 0.8867±0.0067$^{+}$ | 0.8911±0.0067$^{+}$ |
| Myeloma | **0.9155±0.0101** | 0.8971±0.0072$^{+}$ | 0.8825±0.0084$^{+}$ | 0.8450±0.0244$^{+}$ | 0.8702±0.0069$^{+}$ | 0.8550±0.0079$^{+}$ | 0.8522±0.0071$^{+}$ | 0.8304±0.0077$^{+}$ | 0.8186±0.0056$^{+}$ | 0.8535±0.0037$^{+}$ | 0.8529±0.0060$^{+}$ | 0.8594±0.0070$^{+}$ | 0.8482±0.0115$^{+}$ | 0.8425±0.0035$^{+}$ |
| PCMAC | **0.8705±0.0139** | 0.8424±0.0163$^{+}$ | 0.8007±0.0051$^{+}$ | 0.7870±0.0080$^{+}$ | 0.8103±0.0090$^{+}$ | 0.8036±0.0095$^{+}$ | 0.8256±0.0095$^{+}$ | 0.7840±0.0058$^{+}$ | 0.7982±0.0062$^{+}$ | 0.8175±0.0056$^{+}$ | 0.8105±0.0086$^{+}$ | 0.8523±0.0083$^{+}$ | 0.8505±0.0117$^{+}$ | 0.8169±0.0035$^{+}$ |
| Prostate | **0.9701±0.0032** | 0.9595±0.0045$^{+}$ | 0.9638±0.0049$^{+}$ | 0.8458±0.0083$^{+}$ | 0.8887±0.0092$^{+}$ | 0.8721±0.0088$^{+}$ | 0.8711±0.0091$^{+}$ | 0.8526±0.0102$^{+}$ | 0.8485±0.0038$^{+}$ | 0.8840±0.0031$^{+}$ | 0.8839±0.0054$^{+}$ | 0.8704±0.0120$^{+}$ | 0.8703±0.0107$^{+}$ | 0.8743±0.0052$^{+}$ |
| SRBCT | 0.9986±0.0004 | 0.9984±0.0006$^{\approx}$ | 0.9944±0.0043$^{+}$ | 0.9321±0.0085$^{+}$ | 0.9692±0.0148$^{+}$ | 0.9695±0.0091$^{+}$ | 0.9659±0.0048$^{+}$ | 0.9515±0.0116$^{+}$ | 0.9488±0.0050$^{+}$ | **1.0000±0.0000**$^{-}$ | **1.0000±0.0000**$^{-}$ | 0.9853±0.0074$^{+}$ | 0.9829±0.0099$^{+}$ | 0.9829±0.0080$^{+}$ |
| T1D | **0.8598±0.0094** | 0.8232±0.0126$^{+}$ | 0.8246±0.0084$^{+}$ | 0.7679±0.0461$^{+}$ | 0.7610±0.0169$^{+}$ | 0.7318±0.0065$^{+}$ | 0.7237±0.0049$^{+}$ | 0.7113±0.0064$^{+}$ | 0.7063±0.0041$^{+}$ | 0.7256±0.0049$^{+}$ | 0.7227±0.0053$^{+}$ | 0.7118±0.0063$^{+}$ | 0.7088±0.0088$^{+}$ | 0.7179±0.0048$^{+}$ |
| madelon | **0.8822±0.0180** | 0.8659±0.0246$^{\approx}$ | 0.7679±0.0121$^{+}$ | 0.7521±0.0095$^{+}$ | 0.7963±0.0165$^{+}$ | 0.7669±0.0229$^{+}$ | 0.8106±0.0089$^{+}$ | 0.7497±0.0169$^{+}$ | 0.7652±0.0071$^{+}$ | 0.7920±0.0049$^{+}$ | 0.7845±0.0048$^{+}$ | 0.8214±0.0097$^{+}$ | 0.8122±0.0055$^{+}$ | 0.7808±0.0071$^{+}$ |
| Average | **0.9574** | 0.9488 | 0.9325 | 0.8868 | 0.9177 | 0.9021 | 0.8944 | 0.8839 | 0.8698 | 0.9132 | 0.9110 | 0.9065 | 0.9042 | 0.9009 |

*Note*: Bold values indicate the best performance per dataset. Superscripts +, ≈, and − denote that KDS-EL is significantly better, statistically comparable, or significantly worse ($p < 0.05$, two-sided Wilcoxon signed-rank test), respectively.

### Section S4: Granular Per-Dataset Component Contribution (Ablation) Results (Table S4)

Table 9 provides the complete per-dataset performance metrics across the four sequential configurations of the subtractive ablation chain: (i) Standard MEL [1], (ii) MEL+MI (w/o TPS&KER), (iii) KDS-EL$_{\text{KER}}$ (w/o TPS), and (iv) Complete KDS-EL.

Table 8: Comparison of the average number of selected features across 20 benchmark datasets.

| Dataset | KDS-EL | w/o TPS | w/o KER | MEL | FWPSO | PSO-CSM | SaWDE | IGPSO | BPSO | W-MPSO | fMPSO | NSGA-II | MOEA/D | MOPSO |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ALL1 | **39** | 55 | 55 | 4248 | 612 | 1206 | 3667 | 2501 | 5886 | 2333 | 2339 | 5368 | 5548 | 5920 |
| ALL4 | 124 | 152 | **64** | 1082 | 1326 | 1447 | 4581 | 2677 | 6217 | 3645 | 3213 | 5461 | 5736 | 6126 |
| ALL_AML | **67** | 70 | 473 | 2369 | 521 | 949 | 1996 | 1444 | 3266 | 1678 | 1562 | 2872 | 2968 | 3266 |
| Adenoma | 66 | **33** | 120 | 2483 | 278 | 937 | 2102 | 1608 | 3399 | 1691 | 1551 | 3000 | 3127 | 3420 |
| BASEHOCK | **134** | 208 | 812 | 1965 | 2066 | 1113 | 1912 | 1894 | 2419 | 1390 | 1617 | 2073 | 2150 | 2319 |
| COIL20 | 84 | **60** | 348 | 323 | 90 | 167 | 232 | 255 | 407 | 405 | 407 | 283 | 286 | 405 |
| Colon | **14** | 35 | 105 | 702 | 240 | 284 | 621 | 448 | 935 | 580 | 612 | 690 | 730 | 903 |
| DLBCL | 64 | **52** | 615 | 2784 | 876 | 1074 | 2155 | 1718 | 3379 | 1963 | 2067 | 2904 | 3010 | 3349 |
| Gastric | 100 | 97 | **62** | 3604 | 1070 | 2782 | 7072 | 4916 | 11004 | 5744 | 6082 | 10165 | 10342 | 10938 |
| Gastric1 | **31** | 217 | 678 | 7393 | 1420 | 2966 | 6963 | 5074 | 10661 | 6148 | 6044 | 9954 | 10128 | 10706 |
| Gastric2 | **83** | 140 | 444 | 7084 | 1362 | 3062 | 6708 | 5134 | 10627 | 5803 | 5956 | 9919 | 10062 | 10701 |
| ISOLET | **60** | 100 | 225 | 250 | 172 | 194 | 213 | 195 | 272 | 242 | 255 | 185 | 174 | 256 |
| Leukaemia | 66 | **47** | 488 | 2409 | 534 | 934 | 2026 | 1440 | 3246 | 1492 | 1503 | 2893 | 2973 | 3272 |
| Lymphoma | 45 | **23** | 50 | 1404 | 304 | 524 | 1204 | 880 | 1832 | 1027 | 1076 | 1536 | 1609 | 1834 |
| Myeloma | 250 | 365 | **175** | 2137 | 1863 | 1847 | 4543 | 2859 | 6259 | 3854 | 3866 | 5585 | 5839 | 6113 |
| PCMAC | **91** | 329 | 637 | 1339 | 1209 | 651 | 1292 | 1137 | 1635 | 966 | 912 | 1352 | 1431 | 1560 |
| Prostate | 26 | **23** | 27 | 5058 | 1898 | 1541 | 4229 | 2729 | 6211 | 3294 | 3384 | 5532 | 5676 | 6100 |
| SRBCT | 64 | **36** | 129 | 906 | 431 | 366 | 739 | 515 | 1083 | 700 | 742 | 818 | 874 | 1044 |
| T1D | 330 | 366 | **174** | 6715 | 4984 | 9047 | 18009 | 13170 | 26971 | 16096 | 16774 | 25649 | 25939 | 26830 |
| madelon | **18** | 26 | 130 | 190 | 122 | 95 | 162 | 171 | 221 | 149 | 174 | 128 | 132 | 204 |
| Average | **87.8** | 121.7 | 290.6 | 2722.2 | 1068.9 | 1559.3 | 3521.3 | 2538.2 | 5296.5 | 2960.0 | 3006.8 | 4818.4 | 4936.7 | 5263.3 |

*Note*: Bold values indicate the minimum number of selected features per dataset.

Table 9: Granular per-dataset component contribution analysis results across 20 benchmark datasets.

| Dataset | MEL | | MEL+MI | | KDS-EL w/o TPS | | KDS-EL | |
|---|---|---|---|---|---|---|---|---|
| | Acc | Feat | Acc | Feat | Acc | Feat | Acc | Feat |
| ALL1 | 0.9664±0.0023 | 4248 | 0.9996±0.0000 | 55 | 0.9996±0.0000 | 55 | 0.9998±0.0001 | 39 |
| ALL4 | 0.8704±0.0661 | 1082 | 0.9415±0.0112 | 64 | 0.9418±0.0113 | 152 | 0.9522±0.0061 | 124 |
| ALL_AML | 0.9539±0.0008 | 2369 | 0.9886±0.0016 | 473 | 0.9990±0.0002 | 70 | 0.9995±0.0002 | 67 |
| Adenoma | 0.9410±0.0011 | 2483 | 0.9958±0.0081 | 120 | 0.9996±0.0001 | 33 | 0.9996±0.0002 | 66 |
| BASEHOCK | 0.8518±0.0063 | 1965 | 0.8597±0.0062 | 812 | 0.8997±0.0114 | 208 | 0.9113±0.0120 | 134 |
| COIL20 | 0.9634±0.0017 | 323 | 0.9619±0.0007 | 348 | 0.9919±0.0008 | 60 | 0.9939±0.0011 | 84 |
| Colon | 0.8548±0.0276 | 702 | 0.9211±0.0092 | 105 | 0.9347±0.0094 | 35 | 0.9500±0.0075 | 14 |
| DLBCL | 0.9483±0.0058 | 2784 | 0.9902±0.0016 | 615 | 0.9993±0.0004 | 52 | 0.9995±0.0002 | 64 |
| Gastric | 0.8955±0.0786 | 3604 | 0.9748±0.0088 | 62 | 0.9677±0.0093 | 97 | 0.9782±0.0034 | 100 |
| Gastric1 | 0.9310±0.0155 | 7393 | 0.9642±0.0025 | 678 | 0.9738±0.0004 | 217 | 0.9800±0.0003 | 31 |
| Gastric2 | 0.9522±0.0170 | 7084 | 0.9872±0.0038 | 444 | 0.9936±0.0029 | 140 | 0.9981±0.0029 | 83 |
| ISOLET | 0.8501±0.0067 | 250 | 0.8557±0.0063 | 225 | 0.8898±0.0090 | 100 | 0.8894±0.0056 | 60 |
| Leukaemia | 0.9534±0.0015 | 2409 | 0.9895±0.0017 | 488 | 0.9993±0.0002 | 47 | 0.9995±0.0002 | 66 |
| Lymphoma | 0.8731±0.0239 | 1404 | 0.9868±0.0102 | 50 | 0.9994±0.0004 | 23 | 0.9994±0.0003 | 45 |
| Myeloma | 0.8450±0.0244 | 2137 | 0.8825±0.0084 | 175 | 0.8971±0.0072 | 365 | 0.9155±0.0101 | 250 |
| PCMAC | 0.7870±0.0080 | 1339 | 0.8007±0.0051 | 637 | 0.8424±0.0163 | 329 | 0.8705±0.0139 | 91 |
| Prostate | 0.8458±0.0083 | 5058 | 0.9638±0.0049 | 27 | 0.9595±0.0045 | 23 | 0.9701±0.0032 | 26 |
| SRBCT | 0.9321±0.0085 | 906 | 0.9944±0.0043 | 129 | 0.9984±0.0006 | 36 | 0.9986±0.0004 | 64 |
| T1D | 0.7679±0.0461 | 6715 | 0.8246±0.0084 | 174 | 0.8232±0.0126 | 366 | 0.8598±0.0094 | 330 |
| madelon | 0.7521±0.0095 | 190 | 0.7679±0.0121 | 130 | 0.8659±0.0246 | 26 | 0.8822±0.0180 | 18 |

### Section S5: Granular Per-Dataset Computational Wall-Clock Runtime Measurements (Table S5)

Table 10 details the mean wall-clock computational execution time (in seconds, averaged over 10 independent runs) across all 20 benchmark datasets for the 14 compared algorithms.

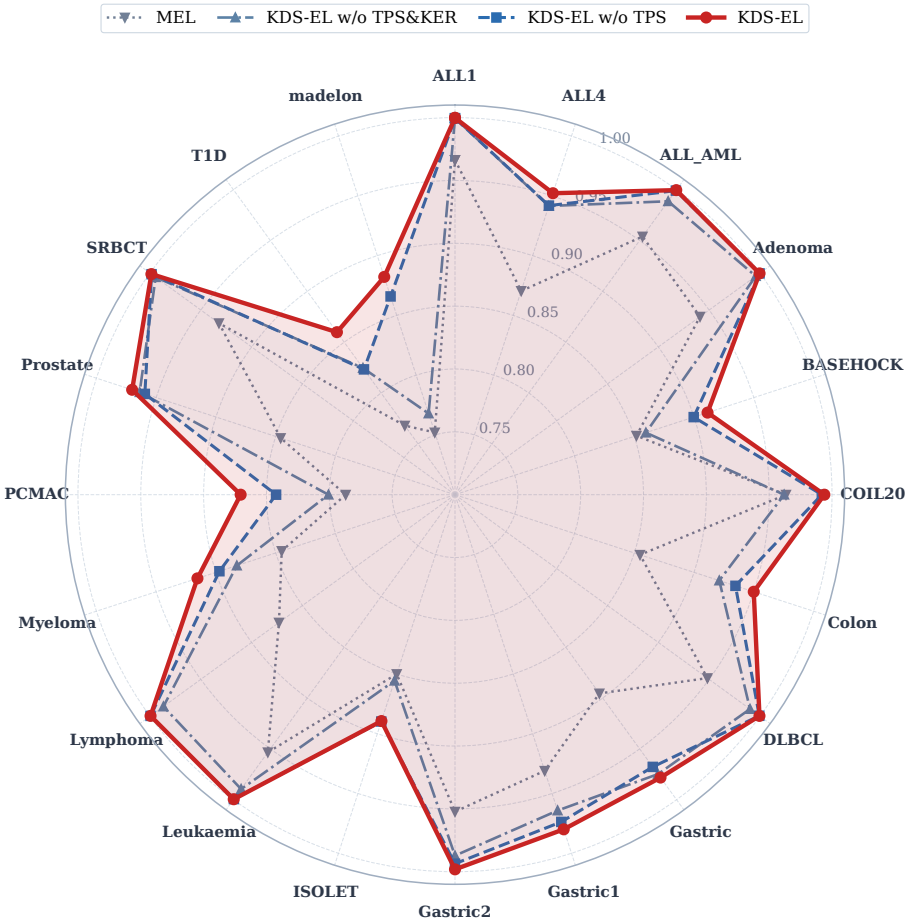



Figure 1: Component contribution heatmap of ablative variants across 20 benchmark datasets.

Table 10: Per-dataset computational runtime in seconds (mean over 10 runs) across 20 benchmark datasets.

| Dataset | KDS-EL | w/o TPS | w/o KER | MEL | FWPSO | PSO-CSM | SaWDE | IGPSO | BPSO | W-MPSO | fMPSO | NSGA-II | MOEA/D | MOPSO |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ALL1 | 40.9 | 39.9 | 39.9 | 176.0 | 19.7 | 82.2 | 67.6 | 91.4 | 101.7 | 82.6 | 86.9 | 96.1 | 98.5 | 102.8 |
| ALL4 | 39.0 | 38.0 | 35.2 | 157.1 | 19.6 | 61.9 | 58.9 | 66.1 | 71.6 | 72.8 | 76.0 | 70.3 | 70.2 | 71.8 |
| ALL_AML | 28.7 | 28.7 | 27.3 | 100.6 | 14.7 | 39.9 | 32.6 | 41.0 | 42.7 | 42.7 | 44.1 | 43.5 | 42.5 | 43.3 |
| Adenoma | 28.3 | 27.7 | 25.7 | 100.8 | 12.9 | 28.6 | 25.6 | 29.4 | 29.9 | 37.5 | 38.0 | 32.1 | 29.5 | 30.3 |
| BASEHOCK | 130.1 | 147.8 | 221.5 | 369.1 | 353.2 | 644.7 | 616.0 | 749.7 | 787.4 | 397.0 | 462.4 | 714.5 | 739.5 | 771.9 |
| COIL20 | 68.3 | 64.3 | 69.7 | 63.2 | 37.7 | 106.0 | 91.6 | 111.6 | 114.2 | 125.4 | 129.6 | 102.6 | 104.3 | 111.8 |
| Colon | 18.1 | 18.5 | 15.6 | 36.3 | 12.2 | 23.0 | 22.8 | 23.1 | 23.5 | 26.7 | 26.8 | 26.3 | 24.2 | 24.8 |
| DLBCL | 28.3 | 28.4 | 27.1 | 98.5 | 15.8 | 40.1 | 36.0 | 42.3 | 44.3 | 46.1 | 48.9 | 44.2 | 43.8 | 44.9 |
| Gastric | 55.8 | 54.2 | 52.1 | 269.3 | 20.4 | 73.0 | 63.2 | 80.6 | 87.3 | 93.2 | 96.8 | 85.5 | 84.9 | 87.4 |
| Gastric1 | 62.9 | 65.7 | 72.7 | 312.0 | 32.4 | 156.7 | 130.9 | 169.7 | 185.6 | 160.3 | 174.8 | 180.5 | 180.9 | 183.7 |
| Gastric2 | 60.8 | 60.6 | 66.1 | 301.2 | 27.2 | 128.3 | 104.5 | 140.5 | 151.0 | 137.5 | 148.3 | 148.0 | 152.1 | 150.3 |
| ISOLET | 59.9 | 61.3 | 55.2 | 50.8 | 41.3 | 88.3 | 82.8 | 91.2 | 94.0 | 98.9 | 101.6 | 87.2 | 84.9 | 90.4 |
| Leukaemia | 28.8 | 28.8 | 27.4 | 98.3 | 14.4 | 39.2 | 32.7 | 40.9 | 42.4 | 41.5 | 43.5 | 43.3 | 43.1 | 43.0 |
| Lymphoma | 21.6 | 21.8 | 19.4 | 59.9 | 12.6 | 24.6 | 24.3 | 25.2 | 25.6 | 30.7 | 31.0 | 28.3 | 26.1 | 26.7 |
| Myeloma | 44.2 | 44.3 | 40.9 | 173.2 | 33.2 | 111.7 | 100.2 | 122.8 | 135.0 | 102.1 | 111.5 | 129.6 | 132.8 | 134.4 |
| PCMAC | 88.2 | 122.5 | 145.8 | 215.9 | 195.5 | 373.1 | 344.5 | 434.7 | 475.3 | 264.7 | 291.2 | 400.0 | 423.2 | 441.0 |
| Prostate | 39.6 | 38.9 | 38.4 | 168.2 | 24.5 | 65.3 | 60.5 | 69.8 | 76.5 | 73.8 | 77.7 | 74.0 | 74.4 | 74.7 |
| SRBCT | 23.0 | 22.4 | 19.5 | 41.2 | 13.4 | 25.3 | 25.1 | 25.6 | 26.0 | 31.8 | 32.3 | 28.8 | 26.8 | 27.4 |
| T1D | 118.8 | 117.4 | 113.2 | 663.6 | 58.3 | 256.6 | 216.1 | 283.6 | 315.2 | 265.7 | 296.1 | 311.4 | 312.2 | 307.8 |
| madelon | 41.5 | 40.4 | 40.1 | 45.3 | 34.1 | 81.4 | 73.7 | 89.0 | 94.2 | 76.5 | 80.4 | 71.8 | 75.8 | 86.9 |

## Section S6: Full Multi-Dataset Parameter Sensitivity Evaluation Grids (Table S6)

Table 11 reports the full multi-dataset one-factor-at-a-time (OFAT) parameter sensitivity evaluation results across the four representative benchmark datasets (*Adenoma*, *Colon*, *ALL4*, and *BASEHOCK*) for each of the five investigated hyper-parameters (WeightDecay, PhaseSplit, flip_frac, MI_init_ratio, and elite_k).

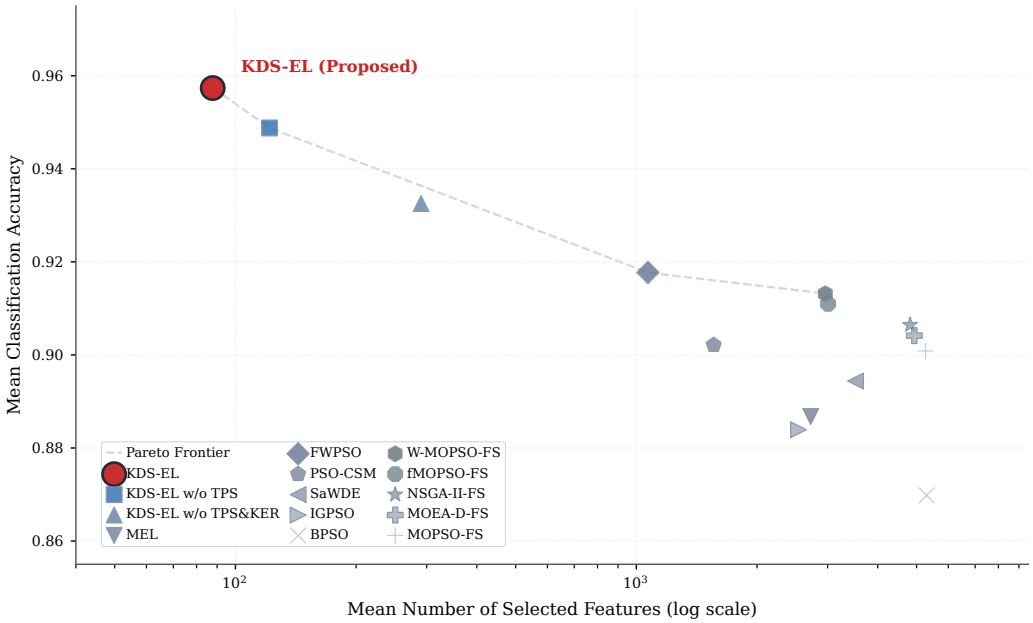



Figure 2: Classification accuracy versus runtime scatter plot for all 14 compared algorithms.

## Section S7: Full Convergence Trajectories and Boxplot Distributions

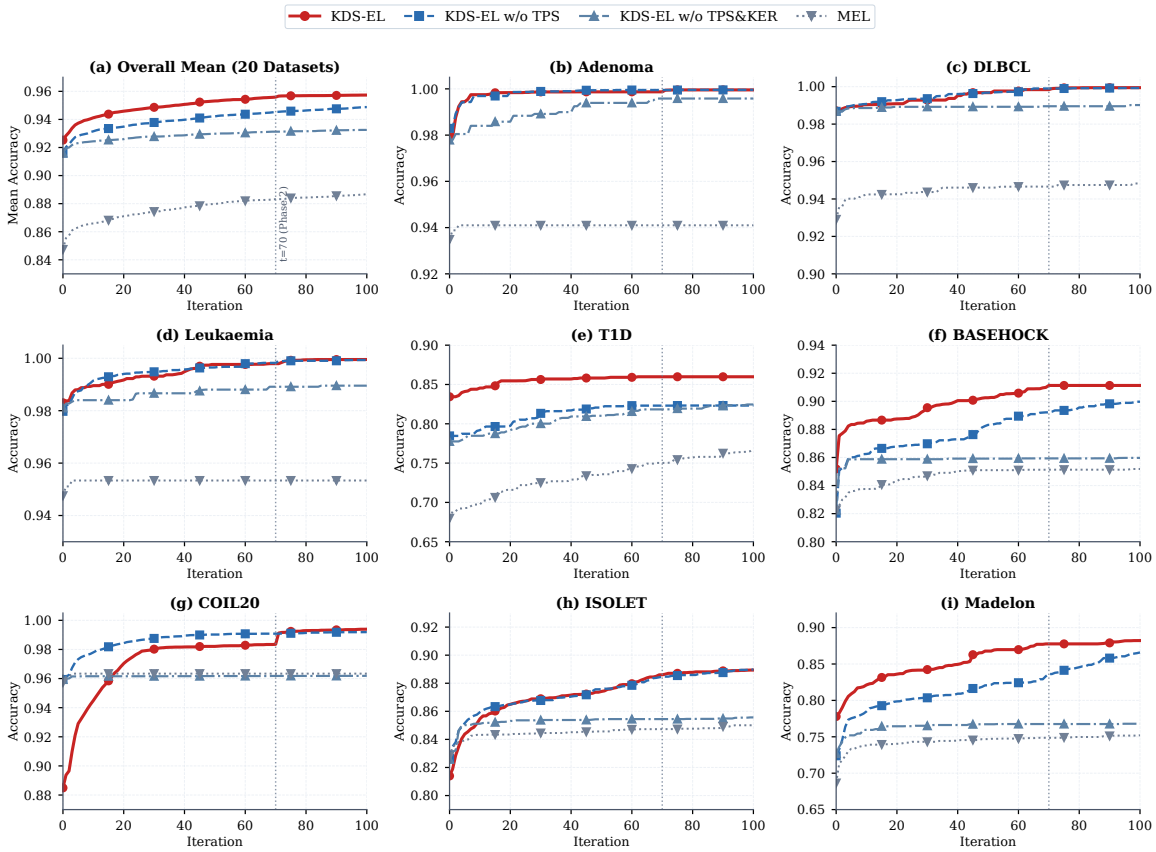



Figure 3: Evolutionary convergence trajectories of all compared algorithms across benchmark datasets.

Table 11: Detailed parameter sensitivity evaluation results across five key hyper-parameters.

| Value | Avg Acc | Avg Feat | Avg Time (s) |
|---|---|---|---|
| *WeightDecay* | | | |
| 0.90 | 0.932 | 158 | 49.2 |
| 0.95 | 0.939 | 127 | 50.8 |
| **0.99 (default)** | **0.944** | **107** | **51.3** |
| 0.995 | 0.943 | 112 | 52.1 |
| 1.00 | 0.941 | 151 | 53.7 |
| *PhaseSplit* | | | |
| 0.3 | 0.940 | 189 | 52.8 |
| 0.5 | 0.942 | 146 | 52.0 |
| 0.6 | 0.943 | 124 | 51.5 |
| **0.7 (default)** | **0.944** | **107** | **51.3** |
| 0.8 | 0.943 | 98 | 50.9 |
| 0.9 | 0.937 | 94 | 50.1 |
| *flip_frac* | | | |
| 0.005 | 0.939 | 96 | 54.8 |
| **0.01 (default)** | **0.944** | **107** | **51.3** |
| 0.02 | 0.942 | 118 | 50.6 |
| 0.05 | 0.937 | 156 | 49.9 |
| 0.10 | 0.931 | 203 | 48.7 |
| *MI_init_ratio* | | | |
| 0.0 | 0.932 | 184 | 62.3 |
| 0.25 | 0.940 | 137 | 55.8 |
| **0.5 (default)** | **0.944** | **107** | **51.3** |
| 0.75 | 0.942 | 101 | 49.6 |
| 1.0 | 0.936 | 96 | 47.8 |
| *elite_k* | | | |
| 1 | 0.938 | 124 | 48.9 |
| 2 | 0.942 | 115 | 50.2 |
| **3 (default)** | **0.944** | **107** | **51.3** |
| 5 | 0.941 | 118 | 52.7 |
| 10 | 0.935 | 141 | 55.9 |

*Note*: Bold entries indicate default parameter configurations.

Table 12: Win counts of pairwise Wilcoxon signed-rank tests ($p < 0.05$) across 20 benchmark datasets.

| Method | KDS-EL | w/o TPS | w/o KER | MEL | FWPSO | PSO-CSM | SaWDE | IGPSO | BPSO | W-MPSO | fMPSO | NSGA-II | MOEA/D | MOPSO |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| KDS-EL | — | 13/20 | 19/20 | 20/20 | 19/20 | 20/20 | 19/20 | 20/20 | 20/20 | 14/20 | 15/20 | 16/20 | 16/20 | 18/20 |
| KDS-EL w/o TPS | **0/20** | — | 15/20 | 20/20 | 20/20 | 20/20 | 19/20 | 20/20 | 20/20 | 14/20 | 15/20 | 15/20 | 15/20 | 17/20 |
| KDS-EL w/o TPS&KER | **0/20** | 0/20 | — | 18/20 | 11/20 | 15/20 | 15/20 | 19/20 | 16/20 | 10/20 | 9/20 | 14/20 | 14/20 | 13/20 |
| MEL | **0/20** | 0/20 | 0/20 | — | 0/20 | 1/20 | 3/20 | 3/20 | 11/20 | 2/20 | 2/20 | 3/20 | 3/20 | 3/20 |
| FWPSO | **0/20** | 0/20 | 0/20 | 0/20 | — | 16/20 | 15/20 | 20/20 | 20/20 | 6/20 | 7/20 | 9/20 | 10/20 | 9/20 |
| PSO-CSM | **0/20** | 0/20 | 0/20 | 0/20 | 0/20 | — | 12/20 | 20/20 | 16/20 | 2/20 | 2/20 | 5/20 | 5/20 | 6/20 |
| SaWDE | **0/20** | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | — | 10/20 | 20/20 | 2/20 | 2/20 | 3/20 | 3/20 | 6/20 |
| IGPSO | **0/20** | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | — | 14/20 | 0/20 | 0/20 | 2/20 | 1/20 | 2/20 |
| BPSO | **0/20** | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | — | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 |
| W-MOPSO-FS | **0/20** | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | — | 2/20 | 10/20 | 12/20 | 12/20 |
| fMOPSO-FS | **0/20** | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | — | 10/20 | 10/20 | 8/20 |
| NSGA-II-FS | **0/20** | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | — | 6/20 | 5/20 |
| MOEA-D-FS | **0/20** | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | — | 4/20 |
| MOPSO-FS | **0/20** | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | 0/20 | — |

*Note*: Entries indicate the number of datasets where the row algorithm significantly outperforms the column algorithm.

## Section S8: Extended Statistical Significance Testing Matrices

Table 12 and Table 13 summarize the pairwise non-parametric Wilcoxon signed-rank test outcomes and the Holm–Bonferroni step-down hypothesis rejection results.

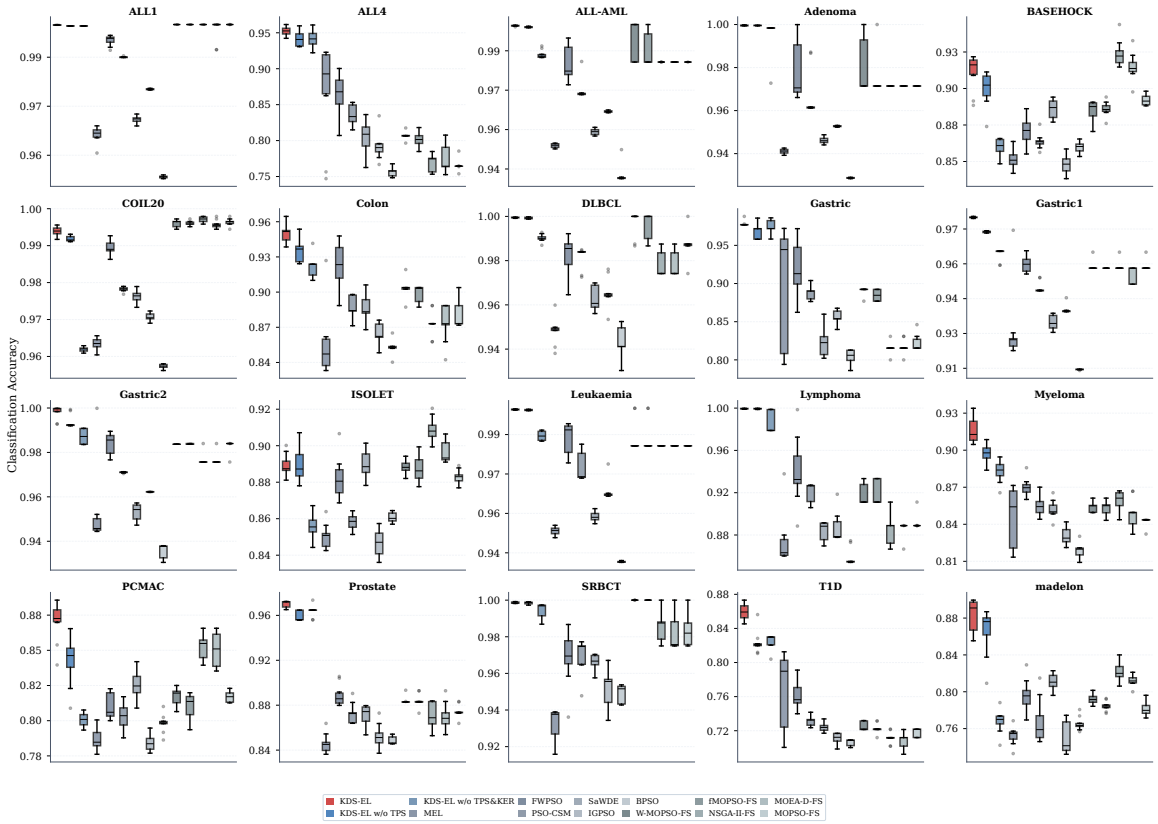



Figure 4: Box plots of classification accuracy across 10 independent runs for all compared algorithms.

Table 13: Holm–Bonferroni corrected pairwise comparisons across 20 benchmark datasets.

| Method | ΔAcc | Combined $p$ | Holm $p$ | Reject $H_0$ |
|---|---|---|---|---|
| KDS-EL w/o TPS | +0.0086 | $< 10^{-15}$ | $< 10^{-15}$ | ✓ |
| KDS-EL w/o TPS&KER | +0.0248 | $< 10^{-15}$ | $< 10^{-15}$ | ✓ |
| MEL | +0.0706 | $< 10^{-15}$ | $< 10^{-15}$ | ✓ |
| FWPSO | +0.0396 | $< 10^{-15}$ | $< 10^{-15}$ | ✓ |
| PSO-CSM | +0.0552 | $< 10^{-15}$ | $< 10^{-15}$ | ✓ |
| SaWDE | +0.0630 | $< 10^{-15}$ | $< 10^{-15}$ | ✓ |
| IGPSO | +0.0734 | $< 10^{-15}$ | $< 10^{-15}$ | ✓ |
| BPSO | +0.0875 | $< 10^{-15}$ | $< 10^{-15}$ | ✓ |
| W-MOPSO-FS | +0.0442 | $< 10^{-15}$ | $< 10^{-15}$ | ✓ |
| fMOPSO-FS | +0.0464 | $< 10^{-15}$ | $< 10^{-15}$ | ✓ |
| NSGA-II-FS | +0.0509 | $< 10^{-15}$ | $< 10^{-15}$ | ✓ |
| MOEA-D-FS | +0.0532 | $< 10^{-15}$ | $< 10^{-15}$ | ✓ |
| MOPSO-FS | +0.0565 | $< 10^{-15}$ | $< 10^{-15}$ | ✓ |

*Note*: ΔAcc denotes the mean accuracy advantage of KDS-EL. Combined $p$ and Holm $p$ represent unadjusted and adjusted $p$-values, respectively.

## References


[1] X. Wang, H. Shangguan, F. Huang, S. Wu, W. Jia, Mel: Efficient multi-task evolutionary learning for high-dimensional feature selection, IEEE Transactions on Knowledge and Data Engineering 36 (8) (2024) 4020–4033. doi:10.1109/TKDE.2024.3366333.

[2] X. Wang, W. Jia, A feature weighting particle swarm optimization method to identify biomarker genes, in: 2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), IEEE, 2022, pp. 830–834. doi:10.1109/BIBM55620.2022.9995376.

[3] B. Wei, S. Yang, W. Zha, L. Deng, J. Huang, X. Su, F. Wang, Particle swarm optimization algorithm based on comprehensive scoring framework for high-dimensional feature selection, Swarm and Evolutionary Computation 95 (2025) 101915. doi:10.1016/j.swevo.2025.101915.

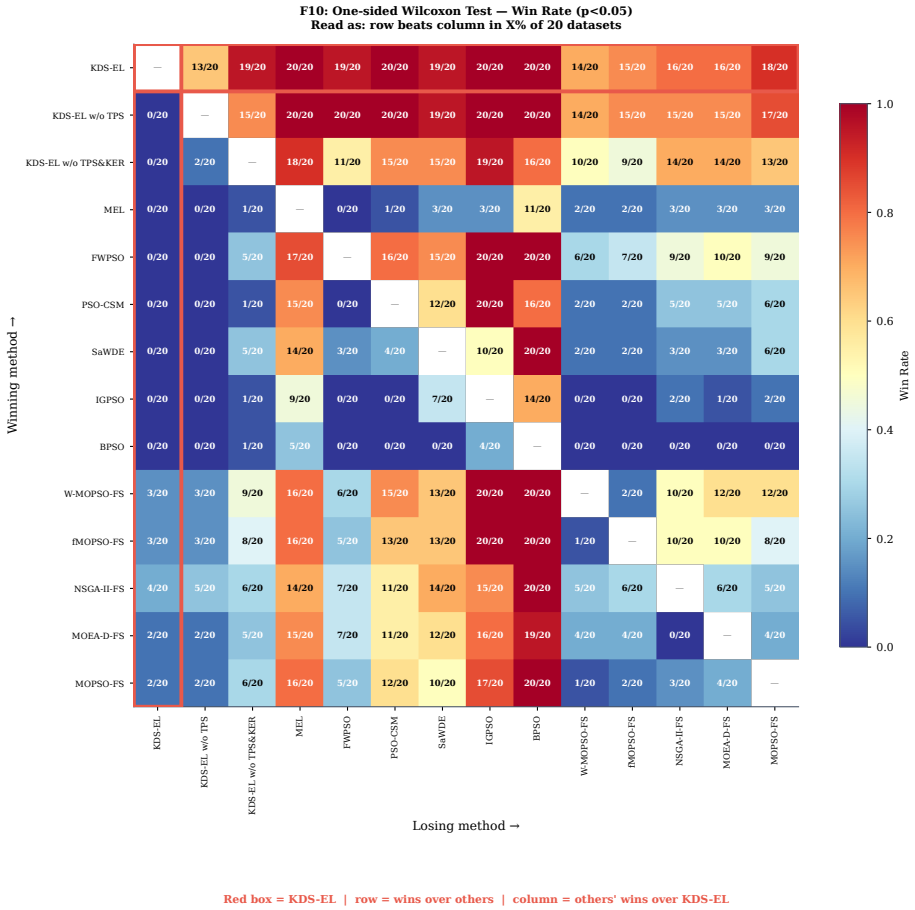



Figure 5: Heatmap of pairwise Wilcoxon signed-rank test $p$-values across all evaluated algorithms.


[4] X. Wang, Y. Wang, K.-C. Wong, X. Li, A self-adaptive weighted differential evolution approach for large-scale feature selection, Knowledge-Based Systems 235 (2022) 107633. doi:10.1016/j.knosys.2021.107633.

[5] Y. Xue, C. Zhang, A novel importance-guided particle swarm optimization based on mlp for solving large-scale feature selection problems, Swarm and Evolutionary Computation 91 (2024) 101760. doi:10.1016/j.swevo.2024.101760.

[6] J. Kennedy, R. C. Eberhart, A discrete binary version of the particle swarm algorithm, Proceedings of the 1997 IEEE International Conference on Systems, Man, and Cybernetics 5 (1997) 4104–4108. doi:10.1109/ICSMC.1997.637339.

[7] F. Han, F. Li, Q. Ling, H. Han, T. Lu, Z. Jiao, H. Zhang, A feature selection method based on feature-label correlation information and self-adaptive MOPSO, Neural Processing Letters 56 (2024) 110. doi:10.1007/s11063-024-11553-9.

[8] K. Deb, A. Pratap, S. Agarwal, T. Meyarivan, A fast and elitist multiobjective genetic algorithm: NSGA-II, IEEE Transactions on Evolutionary Computation 6 (2) (2002) 182–197. doi:10.1109/4235.996017.

[9] Q. Zhang, H. Li, MOEA/D: A multiobjective evolutionary algorithm based on decomposition, IEEE Transactions on Evolutionary Computation 11 (6) (2007) 712–731. doi:10.1109/TEVC.2007.892759.

[10] C. A. C. Coello, G. T. Pulido, M. S. Lechuga, Handling multiple objectives with particle swarm optimization, IEEE Transactions on Evolutionary Computation 8 (3) (2004) 256–279. doi:10.1109/TEVC.2004.826067.